Deployment Issues

This guide covers common deployment-related issues.

Symptoms

- 502 Bad Gateway
- 503 Service Unavailable
- Health check fails
- Frontend shows blank page
- API calls return CORS errors in production

502/503 Errors

Symptom: Reverse proxy returns 502 or 503

Resolution:

1. Verify backend is running:

```
curl http://localhost:5000/health
```

2. Check backend process:

```
# PM2
pm2 status
pm2 logs kuccps-backend

# Docker
docker ps
docker logs kuccps-backend
```

3. Verify Nginx proxy configuration:

```
nginx -t
sudo systemctl reload nginx
```

Health Check Failures

Symptom: `/health` returns non-200 status

Resolution:

1. Check database connectivity
2. Verify `DATABASE_URL` is correct
3. Check for Prisma connection pool exhaustion
4. Review application logs for startup errors

Blank Frontend Page

Symptom: Frontend loads but shows blank page

Resolution:

1. Check browser console for JavaScript errors
2. Verify `VITE_API_URL` is correct
3. Check that `index.html` is served correctly
4. Verify SPA routing configuration in Nginx:

```
location / {
    try_files $uri $uri/ /index.html;
}
```

SSL Certificate Issues

Symptom: `ERR_SSL_PROTOCOL_ERROR` or certificate warnings

Resolution:

1. Verify certificate files exist:

```
ls -la /etc/letsencrypt/live/ticketing.kuccps.ac.ke/
```

2. Check certificate expiry:

```
openssl x509 -in /etc/letsencrypt/live/ticketing.kuccps.ac.ke/cert.pem -noout -
dates
```

3. Renew certificate:

```
sudo certbot renew
sudo systemctl reload nginx
```

Process Management Issues

PM2 Crashes

Resolution:

```
pm2 restart kuccps-backend
pm2 logs kuccps-backend --lines 100
```

Docker Container Exits

Resolution:

```
docker logs kuccps-backend
docker ps -a
```

Check for:

- Missing environment variables
- Port conflicts
- Database connection failures

Cloudflare Tunnel Issues

Symptom: Tunnel connection drops

Resolution:

1. Verify tunnel is running:

```
cloudflared tunnel info kuccps-ticketing
```

2. Check tunnel routes:

```
cloudflared tunnel route dns list
```

3. Restart tunnel:

```
cloudflared tunnel run kuccps-ticketing
```

Environment Variable Issues

Symptom: Application starts but features don't work

Resolution:

1. Compare deployed `.env` with local `.env`
2. Verify no typos in variable names
3. Ensure secrets are properly escaped (no unescaped quotes)
4. Check for duplicate keys in `.env`

Previous
« Authentication Issues

Troubleshooting

Copyright © 2026 KUCCPS. Built with Docusaurus.